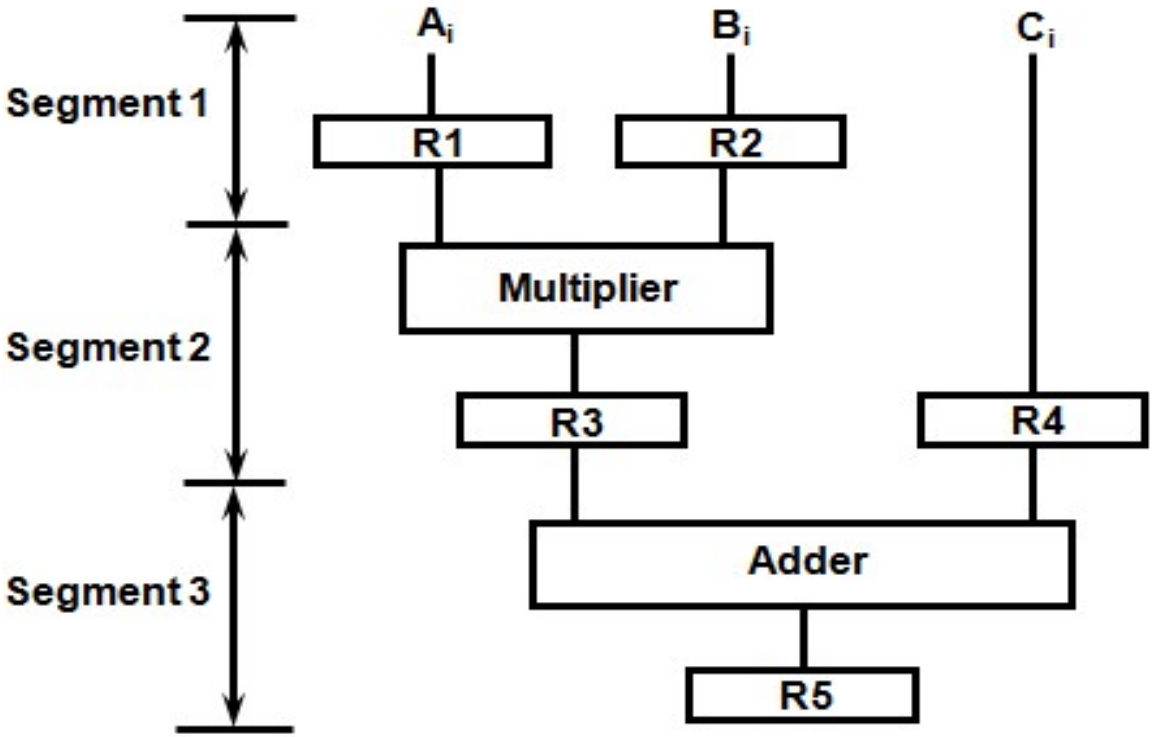
PIPELINING

- Pipelining is a technique of decomposing a sequential process into sub-operations, with each sub-process being executed in a segment that operates concurrently with all other segment.

Example for Pipeline Processing

$A_i * B_i + C_i$ for $i = 1, 2, 3, \ldots, 7$

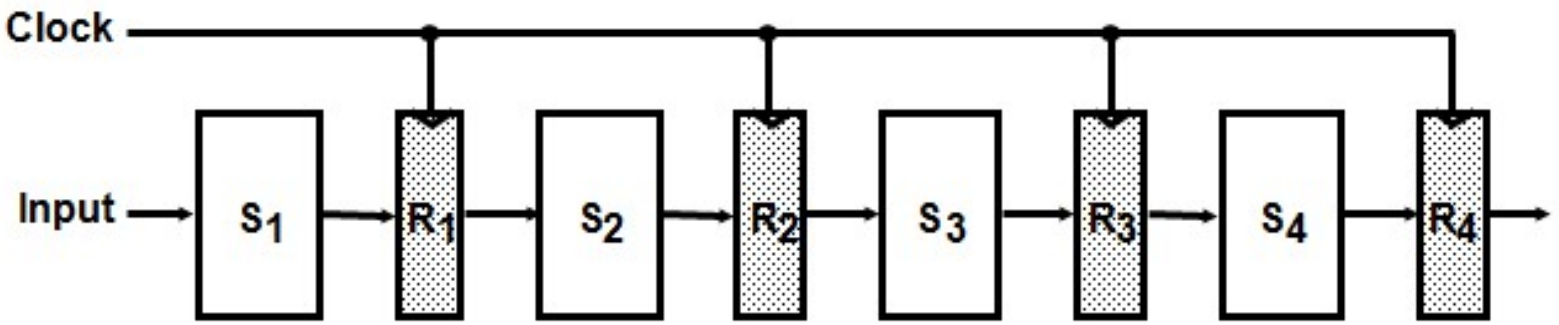


$R1 \leftarrow A_i$, $R2 \leftarrow B_i$	Input A_i and B_i
$R3 \leftarrow R1 * R2$, $R4 \leftarrow C_i$	Multiply and input C_i
$R5 \leftarrow R3 + R4$	Add C_i to product

Operations in each pipeline stage

Clock Pulse Number	Segment 1		Segment 2		Segment 3
	R1	R2	R3	R4	R5
1	A1	B1	---	---	---
2	A2	B2	A1 * B1	C1	---
3	A3	B3	A2 * B2	C2	A1 * B1 + C1
4	A4	B4	A3 * B3	C3	A2 * B2 + C2
5	A5	B5	A4 * B4	C4	A3 * B3 + C3
6	A6	B6	A5 * B5	C5	A4 * B4 + C4
7	A7	B7	A6 * B6	C6	A5 * B5 + C5
8	---	---	A7 * B7	C7	A6 * B6 + C6
9	---	---	---	---	A7 * B7 + C7

General Four-Segment Pipeline



Space-Time Diagram

Segment	1	2	3	4	5	6	7	8	9
1	T1	T2	T3	T4	T5	T6			
2		T1	T2	T3	T4	T5	T6		
3			T1	T2	T3	T4	T5	T6	
4				T1	T2	T3	T4	T5	T6

Clock cycles

Pipeline Speedup

n: Number of tasks to be performed

Pipelined Machine (k segments)
k-segment pipeline with a clock cycle time of t_p is used to execute n tasks

The first task T_1 requires time = $k*t_p$

The remaining (n-1) tasks emerge from the pipe at the rate of one task per clock cycle = $(n-1)* t_p$

To complete n task with k-segment pipeline= k + (n-1) clock cycles

Conventional Machine (Non-Pipelined)
Time required to complete each task = t_n
Time required to complete the n tasks = $n * t_n$

Speedup
S_k: Speedup

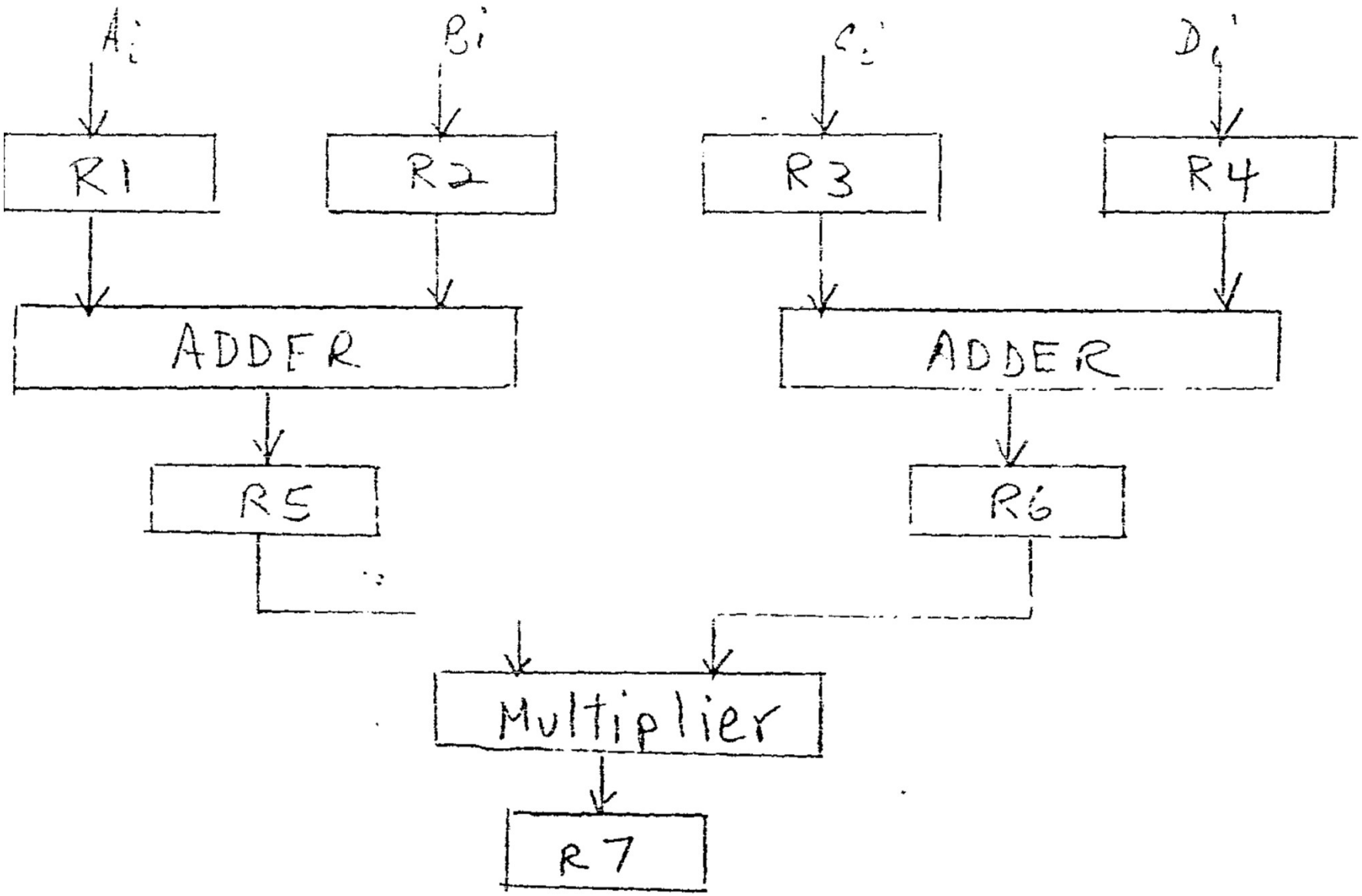
$S_k = n*t_n / (k + n - 1)*t_p$

Q1. In certain scientific computation it is necessary to perform the arithmetic operation (Ai+Bi)*(Ci+Di)

a. Specify the pipeline configuration to carry out this task.
b. List the content of all registers of all the registers in the pipeline for i=1 to 6.

Q2. Draw a space time diagram for a six-segment pipeline showing the time it takes to process eight tasks.

Q3. A non-pipelined system takes 50ns to process a task. The same task can be processed in a six-segment pipeline with a clock cycle of 10ns. Determine the speedup ratio of the pipeline for 100 tasks.


A
B
C
D
R1
R2
R3
R4
ADDER
ADDER
R5
R6
Multiplier
R7

Segment	1	2	3	4	5	6	7	8	9	10	11	12	13
1	T_1	T_2	T_3	T_4	T_5	T_6	T_7	T_8					
2		T_1	T_2	T_3	T_4	T_5	T_6	T_7	T_8				
3			T_1	T_2	T_3	T_4	T_5	T_6	T_7	T_8			
4				T_1	T_2	T_3	T_4	T_5	T_6	T_7	T_8		
5					T_1	T_2	T_3	T_4	T_5	T_6	T_7	T_8	
6						T_1	T_2	T_3	T_4	T_5	T_6	T_7	T_8

$(k-n-1)t_p = 6+8-1 = 13$ cycles

$t_n = 50\ ns$

$k = 6$

$t_p = 10\ ns$

$n = 100$

$$S = \frac{n\,t_n}{(k+n-1)\,t_r} = \frac{100 \times 50}{(6-99)\times 10} = 4.76$$

$$S_{max} = \frac{t_n}{t_p} = \frac{50}{10} = 5$$